

## General Description

The AGM405AP1 combines advanced trench MOSFET technology with a low resistance package to provide extremely low  $R_{DS(ON)}$ .

This device is ideal for load switch and battery protection applications.

## Features

- Advance high cell density Trench technology
- Low  $R_{DS(ON)}$  to minimize conductive loss
- Low Gate Charge for fast switching
- Low Thermal resistance
- 100% Avalanche tested
- 100% DVDS tested

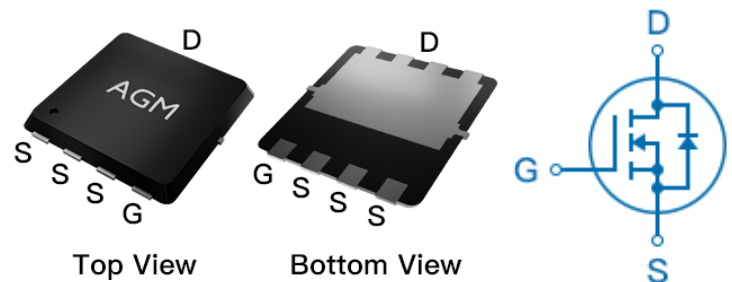
## Application

- MB/VGA Vcore
- SMPS 2<sup>nd</sup> Synchronous Rectifier
- POL application
- BLDC Motor driver

## Product Summary

| BVDSS | RDSON | ID  |
|-------|-------|-----|
| 40V   | 5.7mΩ | 45A |

## PDFN3.3\*3.3 Pin Configuration



## Package Marking and Ordering Information

| Device Marking | Device    | Device Package | Reel Size | Tape width | Quantity |
|----------------|-----------|----------------|-----------|------------|----------|
| AGM405AP1      | AGM405AP1 | PDFN3.3*3.3    | 330mm     | 12mm       | 5000     |

**Table 1. Absolute Maximum Ratings (TA=25°C)**

| Symbol      | Parameter                                         | Value      | Unit |
|-------------|---------------------------------------------------|------------|------|
| VDS         | Drain-Source Voltage (VGS=0V)                     | 40         | V    |
| VGS         | Gate-Source Voltage (VDS=0V)                      | ±20        | V    |
| ID          | Drain Current-Continuous(Tc=25°C) <b>(Note 1)</b> | 45         | A    |
|             | Drain Current-Continuous(Tc=100°C)                | 29         | A    |
| IDM (pluse) | Drain Current-Pulsed <b>(Note 2)</b>              | 62         | A    |
| PD          | Maximum Power Dissipation(Tc=25°C)                | 27         | w    |
|             | Maximum Power Dissipation(Tc=100°C)               | 11         | w    |
| EAS         | Avalanche energy <b>(Note 3)</b>                  | 100        | mJ   |
| TJ,TSTG     | Operating Junction and Storage Temperature Range  | -55 To 150 | °C   |

**Table 2. Thermal Characteristic**

| Symbol | Parameter                                                       | Typ | Max | Unit |
|--------|-----------------------------------------------------------------|-----|-----|------|
| RθJA   | Thermal Resistance Junction-ambient (Steady State) <sup>1</sup> | --- | 60  | °C/W |
| RθJC   | Thermal Resistance Junction-Case <sup>1</sup>                   | --- | 4.5 | °C/W |

**Table 3. Electrical Characteristics (TJ=25℃ unless otherwise noted)**

| Symbol                             | Parameter                        | Conditions                             | Min | Typ  | Max  | Unit |
|------------------------------------|----------------------------------|----------------------------------------|-----|------|------|------|
| On/Off States                      |                                  |                                        |     |      |      |      |
| BVDSS                              | Drain-Source Breakdown Voltage   | VGS=0V ID=250μA                        | 40  | --   | --   | V    |
| IDSS                               | Zero Gate Voltage Drain Current  | VDS=40V,VGS=0V                         | --  | --   | 1    | μA   |
| IGSS                               | Gate-Body Leakage Current        | VGS=±20V,VDS=0V                        | --  | --   | ±100 | nA   |
| VGS(th)                            | Gate Threshold Voltage           | VDS=VGS,ID=250μA                       | 1.2 | 1.6  | 2.2  | V    |
| gFS                                | Forward Transconductance         | VDS=5V,ID=10A                          | --  | 20   | --   | S    |
| RDS(on)                            | Drain-Source On-State Resistance | VGS=10V, ID=15A                        | --  | 5.7  | 8.0  | mΩ   |
|                                    |                                  | VGS=4.5V, ID=10A                       | --  | 9.5  | 14   | mΩ   |
| Dynamic Characteristics            |                                  |                                        |     |      |      |      |
| Ciss                               | Input Capacitance                | VDS=20V,VGS=0V,<br>F=1MHZ              | --  | 685  | --   | pF   |
| Coss                               | Output Capacitance               |                                        | --  | 190  | --   | pF   |
| Crss                               | Reverse Transfer Capacitance     |                                        | --  | 37   | --   | pF   |
| Rg                                 | Gate resistance                  | VGS=0V,<br>VDS=0V,f=1.0MHz             | --  | 1.7  | --   | Ω    |
| Switching Times                    |                                  |                                        |     |      |      |      |
| td(on)                             | Turn-on Delay Time               | VGS=10V,VDS=15V,<br>RI=0.75Ω,RGEN=3.3Ω | --  | 13.9 | --   | nS   |
| tr                                 | Turn-on Rise Time                |                                        | --  | 5.7  | --   | nS   |
| td(off)                            | Turn-Off Delay Time              |                                        | --  | 20   | --   | nS   |
| tf                                 | Turn-Off Fall Time               |                                        | --  | 11   | --   | nS   |
| Qg                                 | Total Gate Charge                | VGS=10V, VDS=25V,<br>ID=12A            | --  | 5.7  | --   | nC   |
| Qgs                                | Gate-Source Charge               |                                        | --  | 3    | --   | nC   |
| Qgd                                | Gate-Drain Charge                |                                        | --  | 1.2  | --   | nC   |
| Source-Drain Diode Characteristics |                                  |                                        |     |      |      |      |
| ISD                                | Source-Drain Current(Body Diode) |                                        | --  | --   | 45   | A    |
| VSD                                | Forward on Voltage               | VGS=0V,IS=15A                          | --  | --   | 1.2  | V    |
| trr                                | Reverse Recovery Time            | IF=15A , dI/dt=100A/μs ,<br>TJ=25℃     | --  | --   | --   | ns   |
| Qrr                                | Reverse Recovery Charge          |                                        | --  | --   | --   | nc   |

Notes 1.The maximum current rating is package limited.

Notes 2.Repetitive Rating: Pulse width limited by maximum junction temperature

Notes 3.EAS condition: TJ=25℃, VDD=25V, Vgs=10V, ID=20A, L=0.5mH, RG=25ohm

## Typical Characteristics

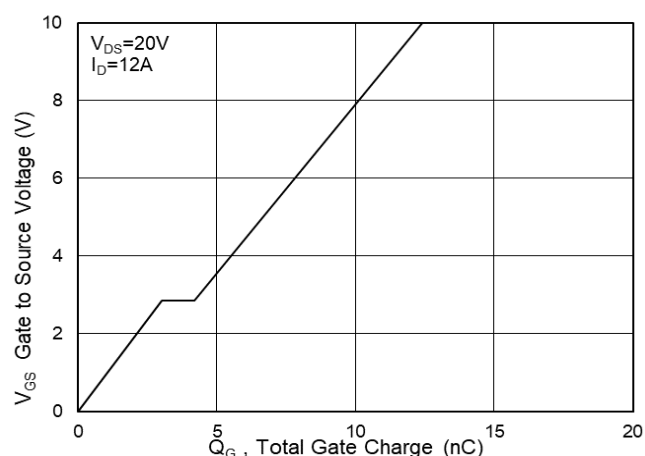
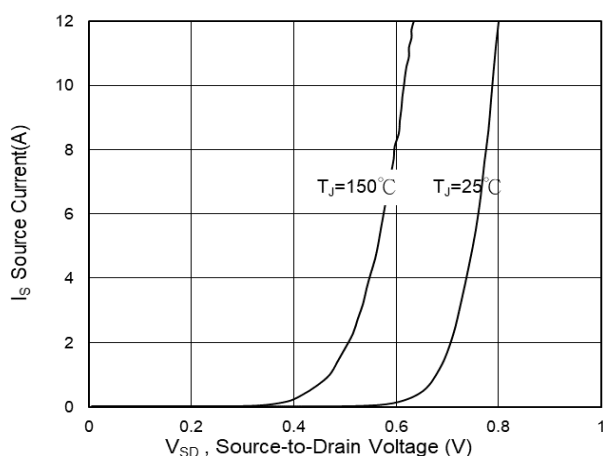
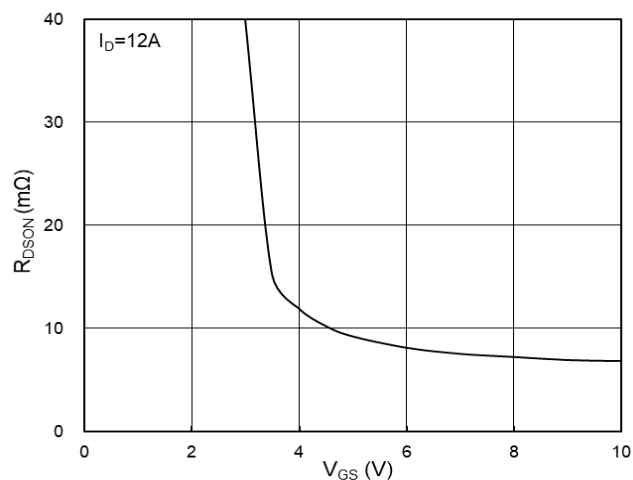
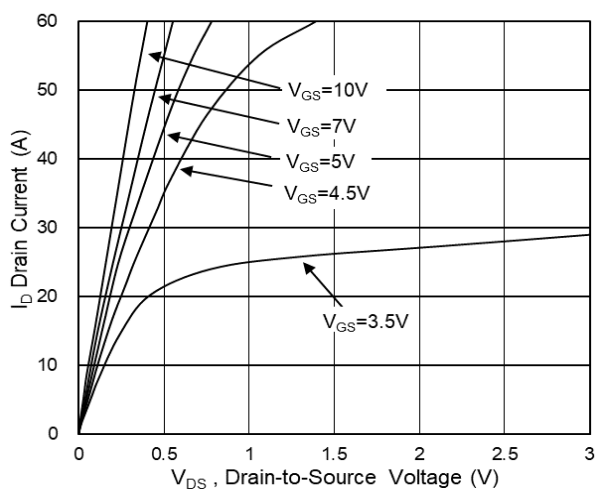


Fig.3 Source Drain Forward Characteristics

Fig.4 Gate-Charge Characteristics

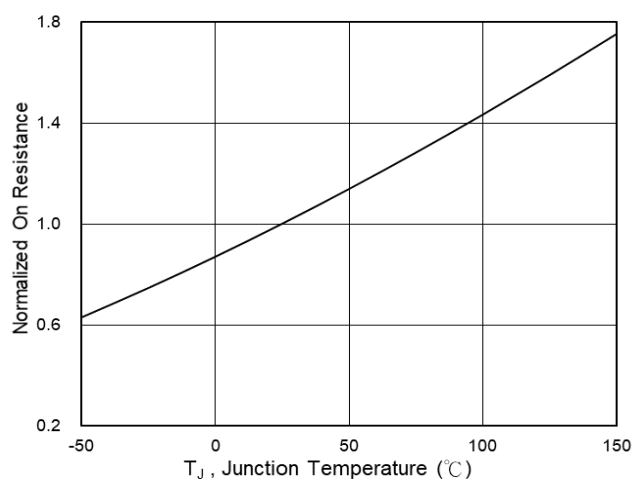
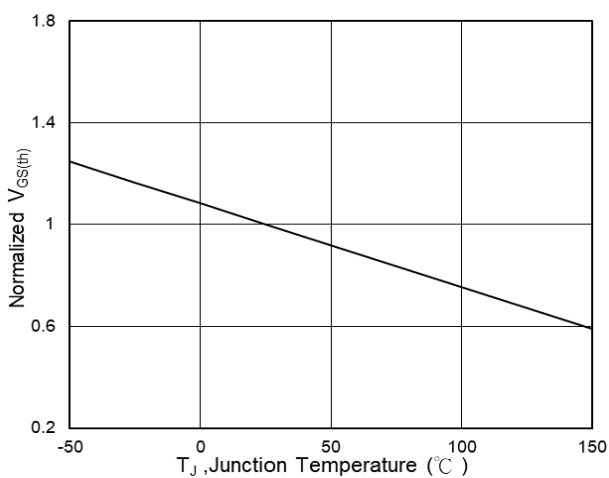


Fig.5 Normalized  $V_{GS(th)}$  vs.  $T_J$

Fig.6 Normalized  $R_{DS(on)}$  vs.  $T_J$

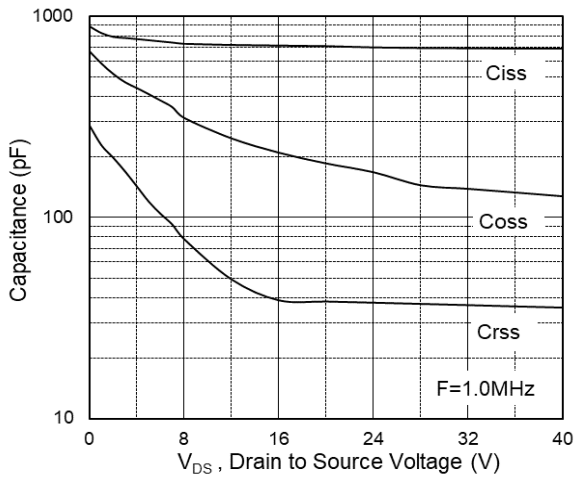


Fig.7 Capacitance

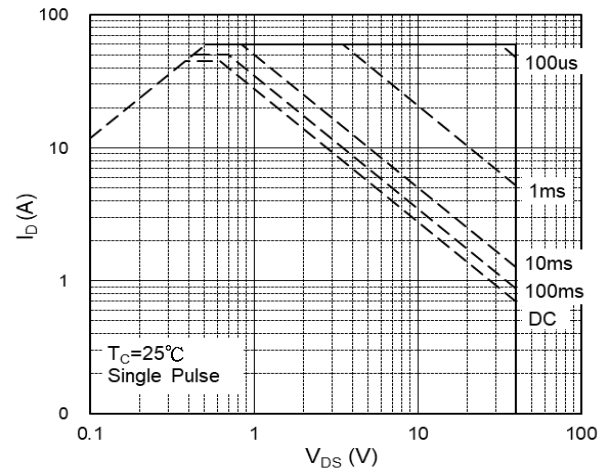


Fig.8 Safe Operating Area

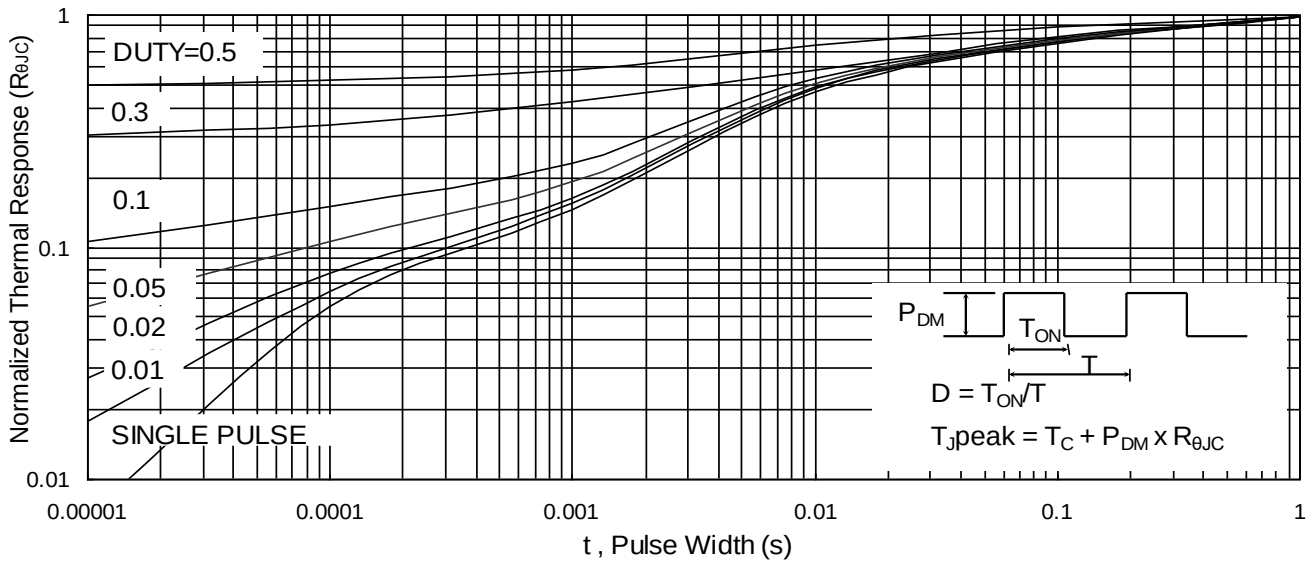


Fig.9 Normalized Maximum Transient Thermal Impedance

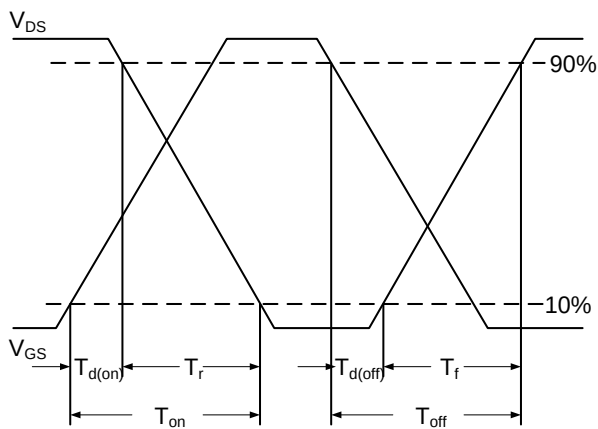


Fig.10 Switching Time Waveform

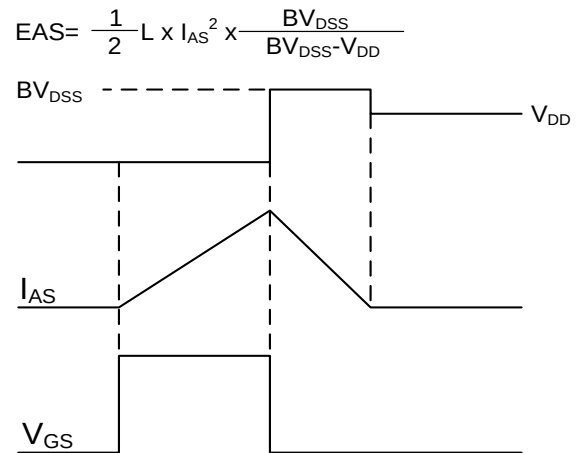
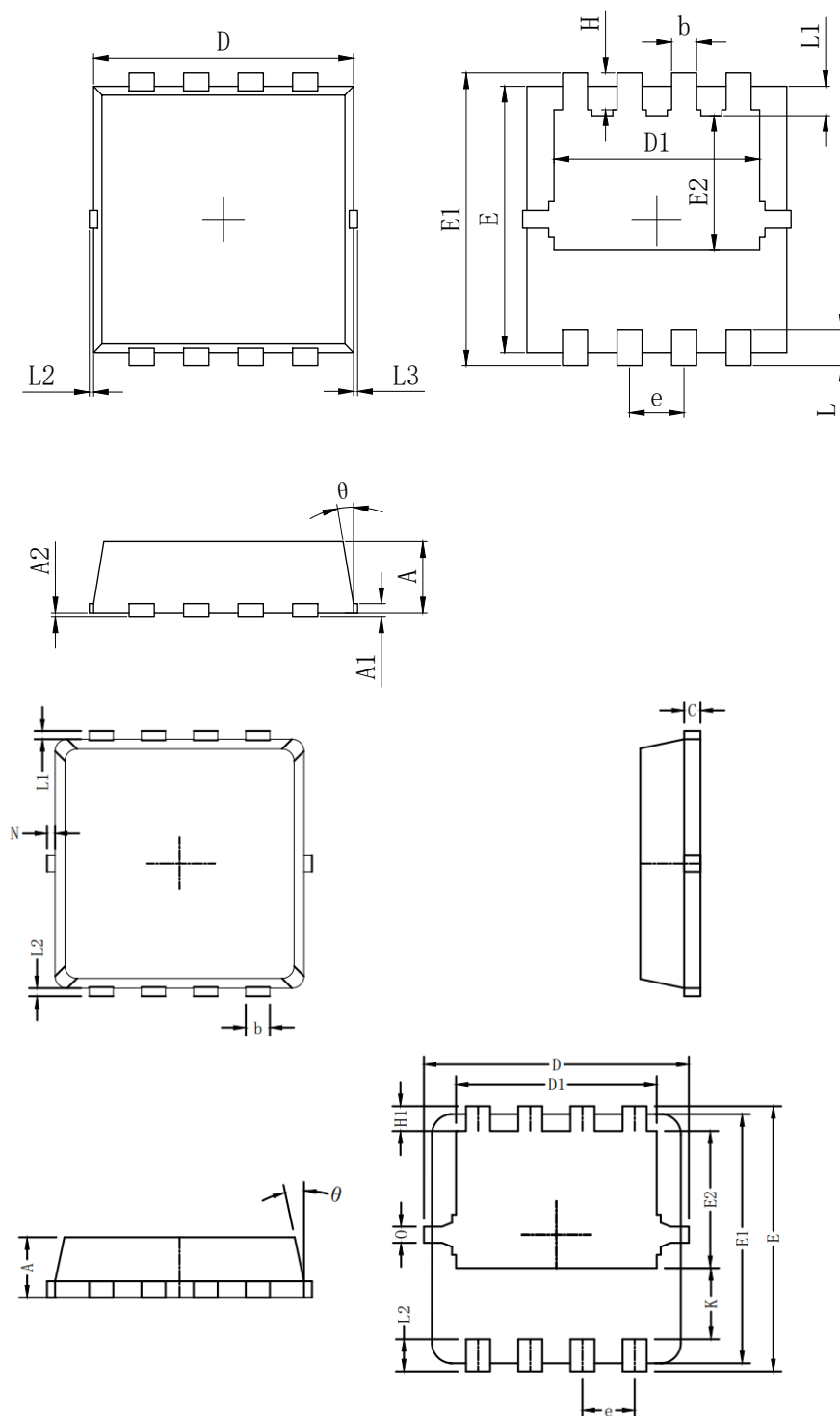


Fig.11 Unclamped Inductive Waveform

# Dimensions (PDFN3.3×3.3)



| SYMBOL   | MILLIMETER |       |       |
|----------|------------|-------|-------|
|          | MIN        | Typ.  | MAX   |
| A        | 0.700      | 0.800 | 0.900 |
| A1       | 0.152 REF. |       |       |
| A2       | 0°0.05     |       |       |
| D        | 3.000      | 3.100 | 3.200 |
| D1       | 2.300      | 2.450 | 2.600 |
| E        | 2.900      | 3.000 | 3.100 |
| E1       | 3.150      | 3.300 | 3.450 |
| E2       | 1.320      | 1.520 | 1.720 |
| b        | 0.200      | 0.300 | 0.400 |
| e        | 0.550      | 0.650 | 0.750 |
| L        | 0.300      | 0.400 | 0.500 |
| L1       | 0.180      | 0.330 | 0.480 |
| L2       | 0°0.100    |       |       |
| L3       | 0°0.100    |       |       |
| H        | 0.315      | 0.415 | 0.515 |
| $\theta$ | 8°         | 10°   | 12°   |

| Symbols  | Millimeters |      |      |
|----------|-------------|------|------|
|          | MIN.        | NOM. | MAX. |
| A        | 0.65        | 0.75 | 0.85 |
| b        | 0.25        | 0.30 | 0.35 |
| C        | 0.15        | 0.20 | 0.25 |
| D        | 3.00        | 3.10 | 3.20 |
| D1       | 2.40        | 2.50 | 2.60 |
| E        | 3.20        | 3.30 | 3.40 |
| E1       | 3.00        | 3.10 | 3.20 |
| E2       | 1.60        | 1.70 | 1.80 |
| e        | 0.65 BSC.   |      |      |
| H1       | 0.21        | 0.31 | 0.41 |
| H2       | 0.30        | 0.40 | 0.50 |
| K        | 0.78        | 0.88 | 0.98 |
| L1/L2    | 0.10 REF.   |      |      |
| $\theta$ | 11°         | 12°  | 13°  |
| N        | 0           | -    | 0.15 |
| 0        | 0.2 REF.    |      |      |

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